

Unequal Job Opportunities and the Measurement of Welfare Inequality

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France, EU-SILC priced through EUROMOD.

Seminar of 17 September. Welfare content is presented under ruling R6; the quantitative decomposition is work in progress.

Question and contribution

Two people with the same tastes and the same wage need not face the same set of jobs.

- How much of measured welfare inequality is attributable to unequal job access and earning opportunities, rather than to preferences?
- What does a model that assumes a common choice set do to that attribution?

Both questions are about a *measure*, not about a policy counterfactual.

The contribution is a structural opportunity set plus a welfare measure defined on it.

- A random-utility, random-opportunity (RURO) model in which *which jobs a household can reach* is estimated, not assumed.
- The Haydar–Maniquet money-metric measure W^1 , applied on the estimated reachable set rather than on a common grid.
- A decomposition architecture that separates preferences from job access, earning opportunities and household endowments and needs.

Third item: the *architecture* is presented today. Its quantitative output is not.

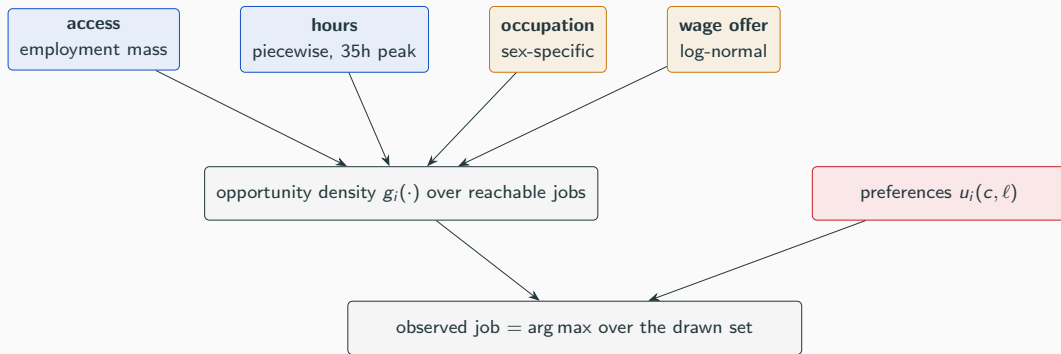
Data and the structural RURO model

France, EU-SILC priced through EUROMOD: 1 540 single-adult and 2 223 couple households.

- Estimation frames: corrected criterion-A panels, S11 model of record.
- Estimation frame: each household's observed job plus 100 drawn alternatives, every one priced through the tax–benefit system.
- Fit diagnostics use a separate, larger panel: 2 048 common quadrature nodes per household.
- Weights are the survey weights dwt throughout.

S11 is the single model of record on these slides. Earlier S8 and R240 frames and their estimates are not shown and are not comparable.

A job is a package; a household draws from its own distribution over packages.



Access and capability are not separately identified; the access factor is their product.

Preferences and the opportunity density are estimated in one likelihood.

$$\Pr(j_i \mid x_i) \propto \exp\{ u_i(c_{ij}, \ell_{ij}) + \log g_i(j) - \log q_i(j) \}$$

u preferences g estimated opportunity density q numerical proposal density, divided out

The proposal correction is what makes the drawn alternatives a simulator for the estimated set rather than the choice set itself.

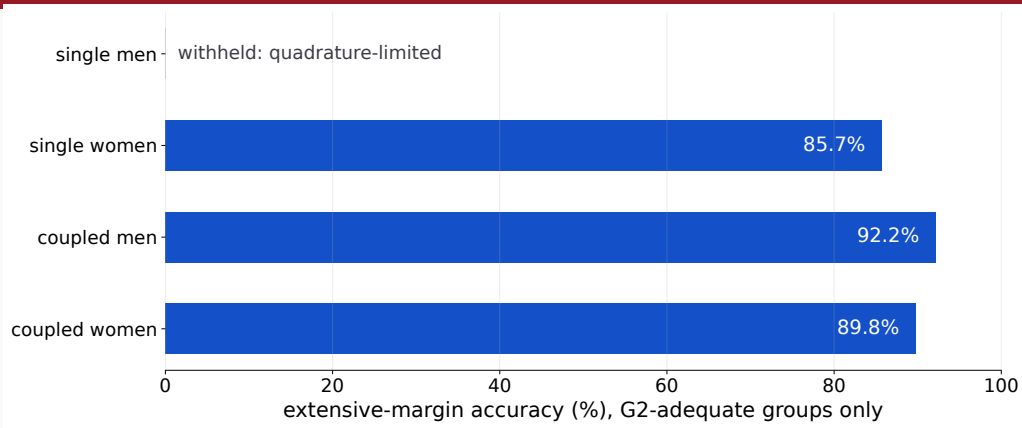
Positive estimates and fit

S11 model of record: 41 free parameters for singles, 47 for couples.

	singles	couples
free parameters	41	47
estimation households	1 540	2 223
objective at $\hat{\theta}$	6253.463	10283.034

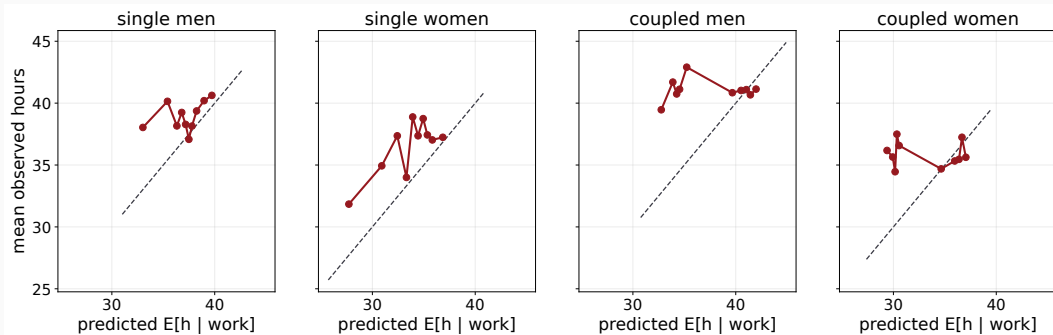
Objectives reproduce exactly under the independent diagnostics evaluator, package commit 55bb0d0.

Extensive-margin accuracy, shown only for the groups where it passes the quadrature-adequacy gate.



Aggregate participation and occupation margins are reproduced reasonably closely, while hours fit is uneven; the group-level fit classification is not used as a headline result. This slide's gate is narrower: G2 (weighted), Monte Carlo s.e. $\leq 0.25 \times$ the sampling s.d., applied to the extensive-accuracy statistic only. 3 of four groups clear that specific gate; 1 is quadrature-limited and withheld, not reported at a worse label. Source: POSFIT v3 (MNL_posfit, branch diagnostics/posfit-v3, commit 96693269); same extensive-margin numbers as v2b (commit

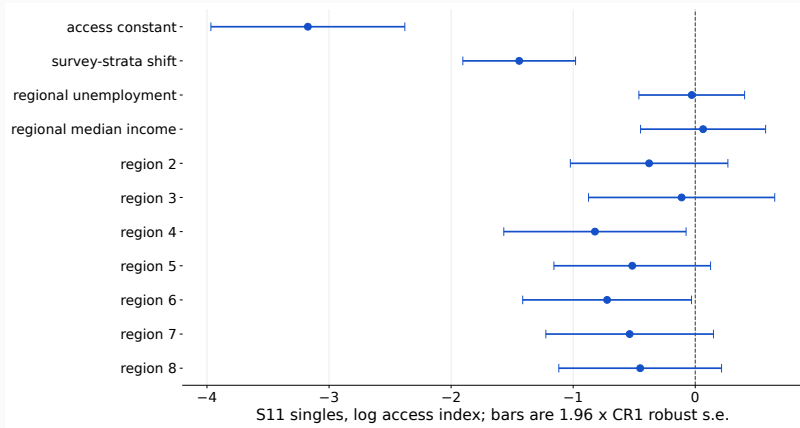
Calibration conditioned on prediction: observed hours against predicted hours, by predicted decile.



support coverage audited (POSFIT v2b); calibration statistics partly quadrature-limited; fit verdicts open pending Deputy review

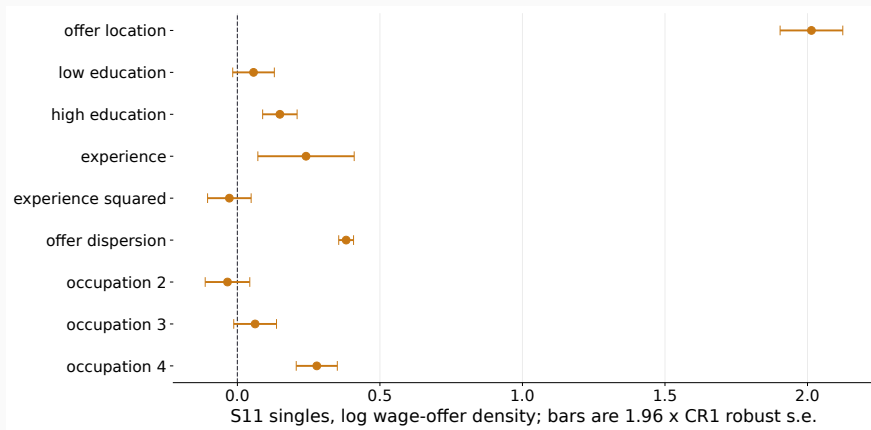
Evidence for heterogeneous opportunities

The access kernel is estimated, and it varies across households.



S11 singles; horizontal bars are $1.96 \times \text{CR1}$ cluster-robust standard errors. Structural and model-conditional; not causal.

So is the earning-opportunity kernel: an offer distribution, not a single wage.



S11 singles; occupation terms shift the offer location. Dispersion $\hat{\sigma} = 0.382$ ($z = 28.7$).

The welfare measure

Haydar–Maniquet W^1 : the consumption at home that would leave the household exactly as well off.

$$W_i^1 = m_i(o; z_i) \quad \text{where} \quad u_i(m_i(o), o) = u_i(z_i)$$

z_i attained bundle o the non-employment state m_i the money metric

Mapping F is the primary empirical implementation (ruling R1). Mapping M is sensitivity only and requires an unidentified intensity parameter; it is not in the baseline.

Under the current specification W^1 coincides with the staying-home equivalent — as a fact about this domain.

$$W_{i,F}^1 = C_i^{\text{obs}} \exp \left[\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right]$$

- Inputs: C_i^{obs} , $L_i(j_i^{\text{obs}})$, $L_i(o)$, β_c — nothing else.
- No opportunity density g , no proposal q , no shock ε , no intensity κ .

This is an **empirical-domain coincidence** under log consumption and universally available non-employment. It is **not** the theorem $W^1 = W^4$ in the Haydar–Maniquet theory.

Equivalised W^1 -F distribution (primary)

Baseline W^1 -F, equivalised, single adults.

	C^{eq}	W_F^{1eq}
households	1 540	1 540
weighted mean	1 766	1 302
weighted median	1 588	1 165
weighted Gini	0.263292	0.249807

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: the ratified modified-OECD scale (Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, s1; JMP_SCALE_REVIEW_1.equivalence_scale_economics.v1.md). Singles and couples equivalised levels are never compared.

Source: docs/results/ JMP_BASELINE_F1.equivalised_reporting.v1.md, commit 4c4e07e, over the verified construction (MNL commit 6048c9f, independently verified b5550af). This is a descriptive dispersion comparison (C^{eq} against W_F^{1eq} within the singles sample), not a welfare-loss index.

Baseline W^1 -F, equivalised, couples.

	C^{eq}	$W_F^{1\text{eq}}$
households	2 223	2 223
weighted mean	2 245	1 311
weighted median	2 063	1 237
weighted Gini	0.226805	0.197403

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: the ratified modified-OECD scale (Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, s1; JMP_SCALE_REVIEW_1_equivalence_scale_economics_v1.md). Singles and couples equivalised levels are never compared.

Reported separately from singles: no pooled figure, and no cross-sample level comparison is drawn on this slide or elsewhere in this deck. Source: same memo and commits as the singles equivalised slide.

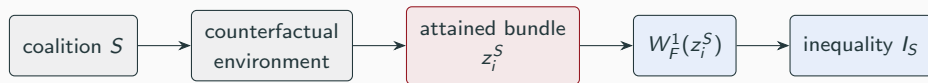
Decomposition architecture

Four operators, one at a time: systematic utility heterogeneity, job access, earning opportunities, household endowments and needs.

operator	replaces, for every household
P	its systematic utility heterogeneity (tastes + reduced-form time constraints) by the common reference
A	its job access by the common reference access
B	its earning opportunities by the common reference offer distribution
D	its household endowments and needs by the common reference budget

Operators only. No share, no magnitude, no ordering is claimed on this slide or anywhere in this deck. Final decomposition semantics are work in progress.

The open problem: the baseline is evaluated at the observed bundle, but an operator moves it.



this step is the unresolved one

Two candidate estimands are under design (R5): a realised-bundle route with conditioned latent heterogeneity, and an ex-ante g -computation route. Neither is executed.

Work in progress

Where the quantitative welfare decomposition stands.

- **Settled.** The measure, its reference domain, the measure-correspondence audit, and the descriptive baseline just shown.
- **In design (R5).** The counterfactual attainment operator, and the choice between the two estimands.
- **Open.** The binding positive-fit verdicts: support coverage is audited (POSFIT v2b), Deputy review of the verdicts is pending.
- **Withdrawn.** Every welfare share previously reported on this project's slides. They were computed on a different functional.

No P/A/B/D share is presented in this deck. R6 permits shares only once the revised F implementation has been executed and independently checked; it has not been.

What I would like from you today.

- Is Mapping F the reference you would want for this question?
- Which counterfactual attainment estimand would you defend?
- What would convince you the access kernel is identified, given that access and capability are not separated?

Thank you.

B1 — Baseline W^1 -F, unequivalised, single adults (verification backup).

households	1 540
weighted mean	1 435
weighted median	1 320
weighted Gini	0.238
observed workers	1 336
observed non-workers	204

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised singles slide (§6) is primary. Source:

MNL/outputs/welfare/baseline_f1_v1/ baseline_f1_full_sample_report_v1.md, commit 6048c9f, independently verified b5550af.

B2 — Baseline W^1 -F, unequivalised, couples (verification backup).

households	2 223
weighted mean	2 496
weighted median	2 356
weighted Gini	0.210
observed workers	2 173
observed non-workers	50

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised couples slide (§6) is primary. Reported separately from singles. No pooled figure: the two samples are separate estimation frames and the welfare unit differs.

B3 — Baseline W^1 -F pre-registered checks

check	singles	couples
C1 non-workers, $\max W^1 - C^{\text{obs}} $	0	0
C2 workers, ratio outside $(0, 1)$	0	0
C3 dependency test $(g, q, \varepsilon, \kappa)$	PASS	PASS
C4 MEASURE-MAP-1R worked households	PASS	PASS
C5 indifference identity, $\max \Delta u $	1.8e-15	1.8e-15
C6 guards triggered	0	0

All six checks were pre-registered in the BASELINE-F-1 ruling before execution. C2 violations were to be explained, never clipped; none occurred.

B4 — The G2 quadrature-adequacy gate

group	extensive accuracy	G2 label
coupled women	89.8%	ADEQUATE (ratio 0.02)
single women	85.7%	ADEQUATE (ratio 0.15)
coupled men	92.2%	ADEQUATE (ratio 0.24)
single men	—	QUADRATURE-LIMITED, withheld

Weighted, all households. Ratio is Monte Carlo s.e. over sampling s.d.; the gate is ≤ 0.25 . Node bootstrap: 200 replicates over 2048 common quadrature nodes per household (diagnostic panel, not the estimation draws). Simulated bands: 500 outcome vectors at fixed $\hat{\theta}$. POSFIT v2b, commit a2e80a8.

B5 — Authority for what is and is not on these slides

ruling	what it settles here
R1	Mapping F primary; $W^1 = W^4$ is a domain coincidence
R2	W1-EA retired from the baseline; preserved, not repaired
R4	MEASURE-MAP-1R: the measure-correspondence gate
R5	counterfactual attainment: in design, not executed
R6	what this seminar may present
E1	BASELINE-F-1 authorised, executed, independently verified
E2	C^{obs} is the estimated utility's own argument
E3	C^{obs} benchmark: not authorised, not produced
E3-EQ	equivalised reporting (§6) primary; modified-OECD ratified (SCALE CLOSED, s1)

R1–R6: docs/normative/JMP_W1_fork_ruling_v1.md, Appendix A. E1–E3: docs/normative/JMP_W1_BASELINE_F1_authorization_and_Cobs_ruling_v1.md. E3-EQ: docs/results/JMP_BASELINE_F1_equivalised_reporting_v1.md, commit 4c4e07e.